People's Democratic Republic of Algeria
Ministry of Higher Education and Scientific Research
BADJI Mokhtar University Annaba
Faculty of Medicine
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Foot joints
& Talocrural joint

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Course outline

- **Introduction: "The joints of the foot"**
- **Talocrural joint :**
  - **Articular surfaces**
  - **Means of union**
  - **Synovial**
  - **Functional Anatomy**

Introduction: "The joints of the foot"

The joints of the foot are divided into three classes:

a. **Talocrural joint :**
   It is a diarthrosis, of the ginglyme type, it connects the two bones of the leg to the talus.
   It is a recessed joint, ensuring the stability of the foot during movements (walking and running).

b. **Tarsal joints :**
   They include:
   - Subtalar joint;
   - Transverse joint of the foot;
   - Articulation of the distal tarsal bones with each other.

c. **Forefoot joints :**
   They include:
   - Tarsometatarsal joints ;
   - Intermetatarsals joints ;
   - Metatarsophalangeal joints ;
   - Interphalangeal joints of the foot.

All these joints allow the movements of the foot: flexion, extension, adduction, abduction, supination, pronation, inversion and eversion.

Talocrural joint

1. Articular surfaces

There are two of them:

a. Tibiofibular articular surfaces :

It is made up of:

- Inferior articular surface of the tibia
- Articular facet of the medial malleolus
- Articular facet of the lateral malleolus

b. Articular surfaces of the trochlea of the talus:

It is a large, protruding joint. It has three surfaces:

- Upper surface of the trochlea: articular with the lower surface of the tibia;
- medial malleolar surface: comma-shaped, articular with the medial malleolus;
- Lateral malleolar surface: triangular, articular with the lateral malleolus;

How are all these articular surfaces organized ? The tibiofibular surfaces form a mortise into which the trochlea of the talus fits.

2. Means of union

a. Joint capsule:

It attaches to the periphery of the joint surfaces.

b. Passive ligaments :

- **Medial ligament:**

  Also called the deltoid ligament. It is strong and triangular, its apex attaching to the medial malleolus. It consists of two layers:

  Surface layer: includes from front to back:
  - Tibionavicular bundle
  - Tibio-calcaneal bundle

  Deep layer: includes from front to back:
  - Anterior tibiotalar bundle
  - Posterior tibiotalar bundle

- **Lateral ligament:**

  It extends from the lateral malleolus and terminates in three bundles:

  Anterior talofibular bundle
  Calcaneofibular bundle
  Posterior talofibular bundle

  Noticed: The lateral ligament is known as "the ankle sprain ligament".

3. Synovial

It lines the inner surface of the capsule.
It forms a cul-de-sac extension between the tibia and the fibula.

4. Functional Anatomy

The talocrural joint has only one degree of freedom, allowing flexion (or dorsiflexion, or elevation) and extension (or plantar flexion, or lowering) of the foot.
These movements occur around a transverse axis.

- Flexion: brings the dorsal side of the foot to the anterior side of the leg, his amplitude is from 20° to 30°.
- Extension: moves the dorsal surface of the foot away from the anterior surface of the leg, his amplitude is from 30° to 60°.

Les articulations du pied

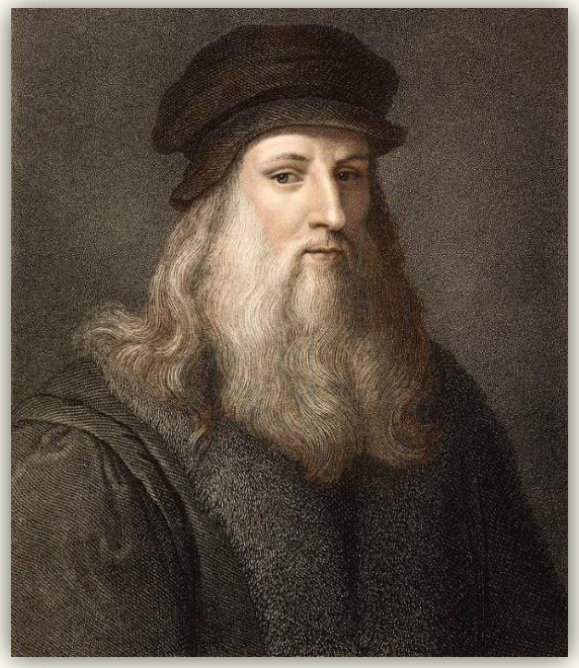


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Le pied par *Léonard de Vinci*





(1452-1519)

Plan du cours

- **Introduction:**

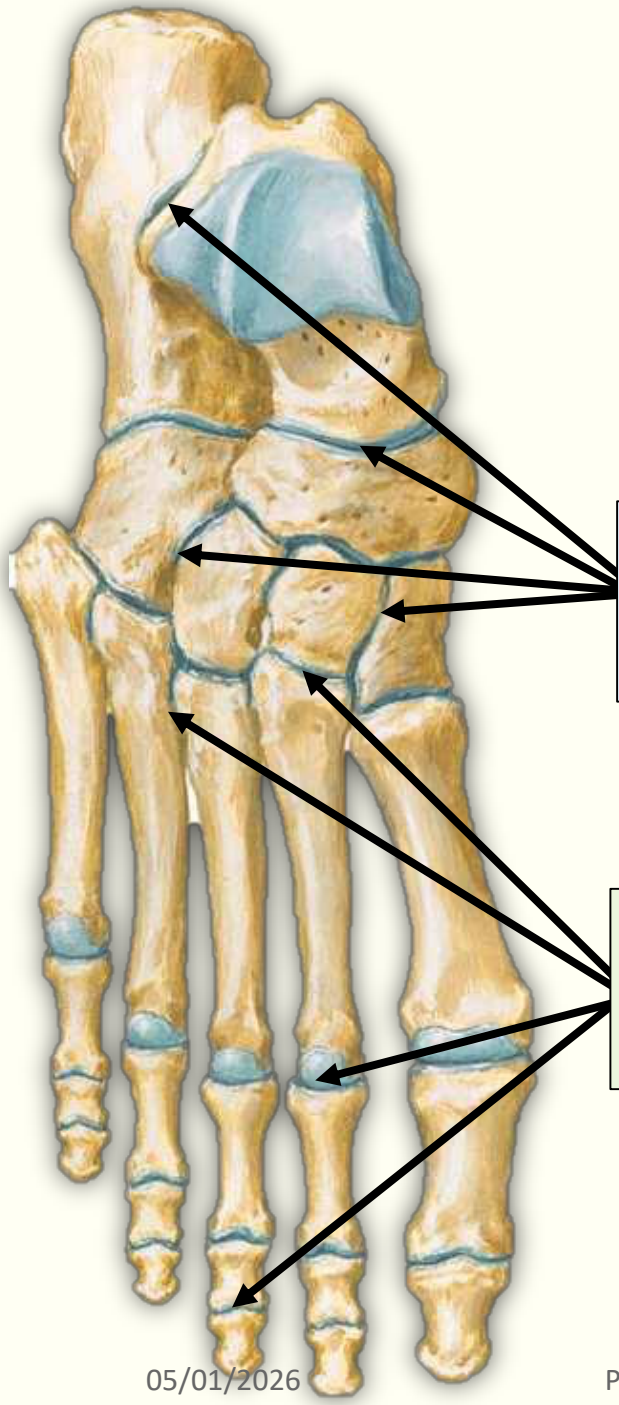
  « Les articulations du pied »

- **Articulation talo-crurale:**
  - Surfaces articulaires
  - Moyens d'union
  - Synoviale
  - Anatomie fonctionnelle



Introduction

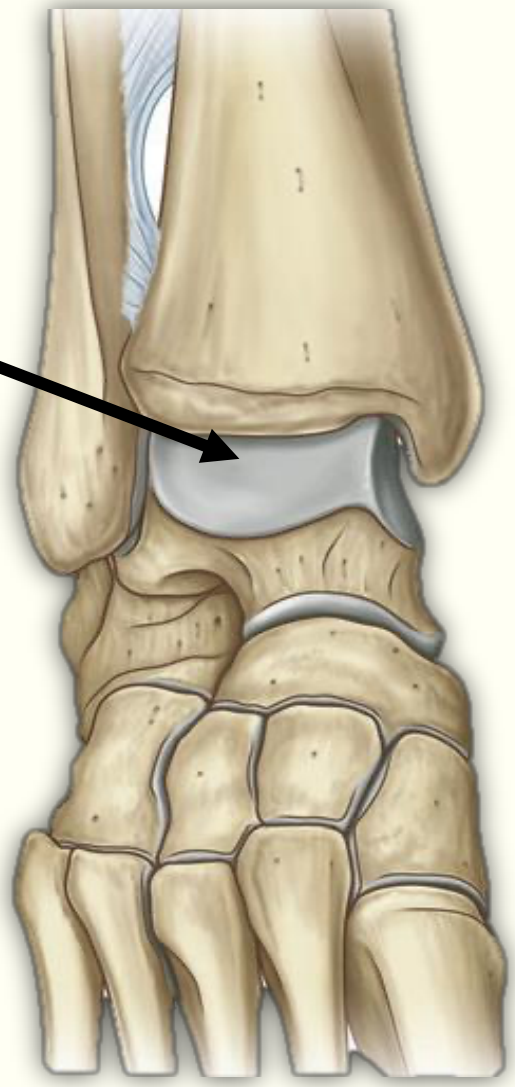
« Les articulations du pied »

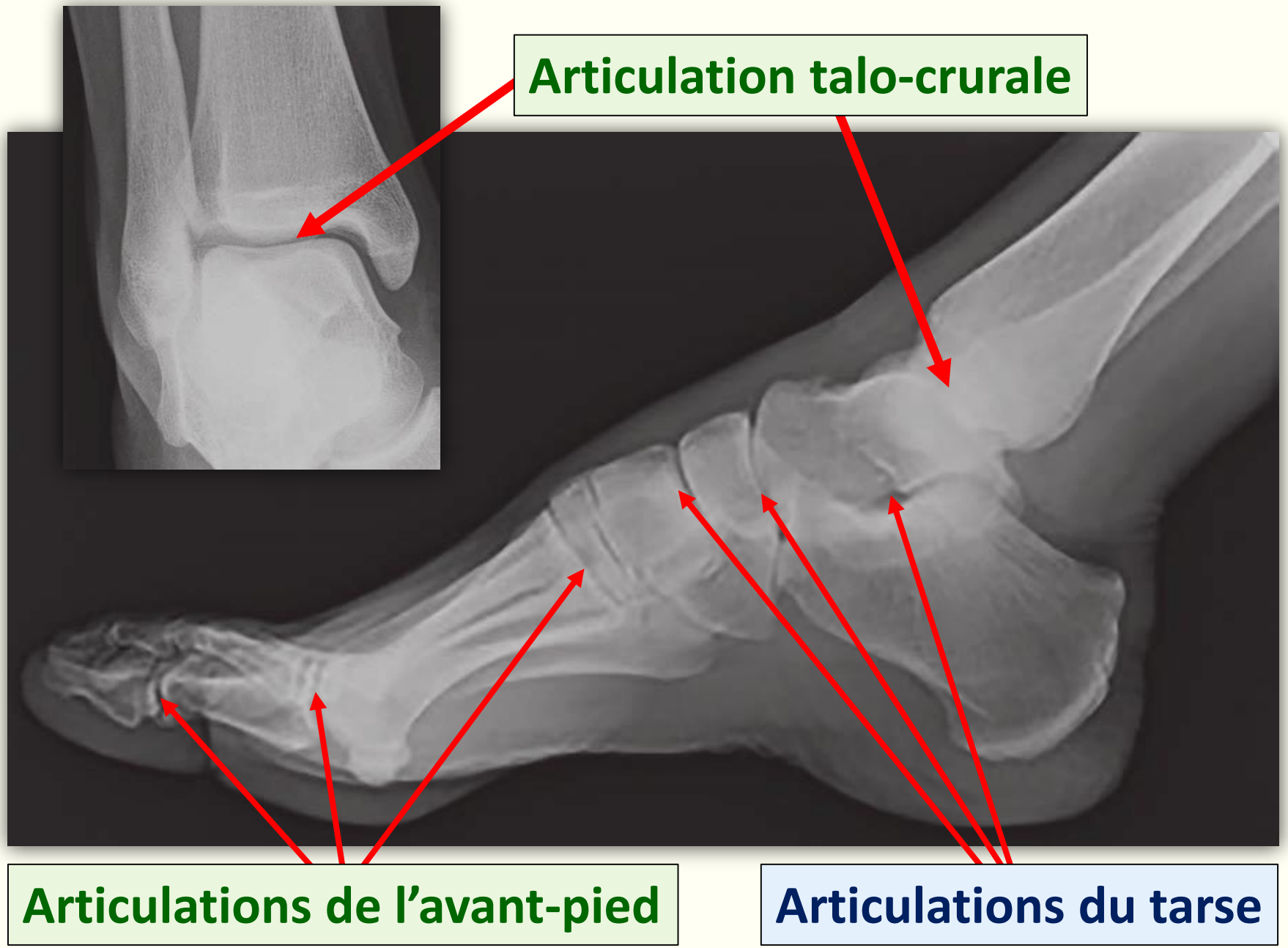


Articulations du tarse

Articulations de l’avant-pied

Articulation talo-crurale



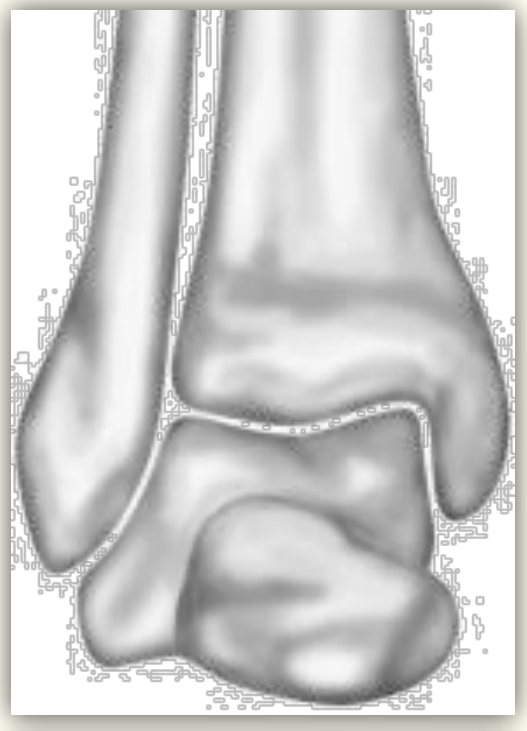

Articulation talo-crurale
Articulations de l’avant-pied
Articulations du tarse

(1)
Articulation talo-crurale



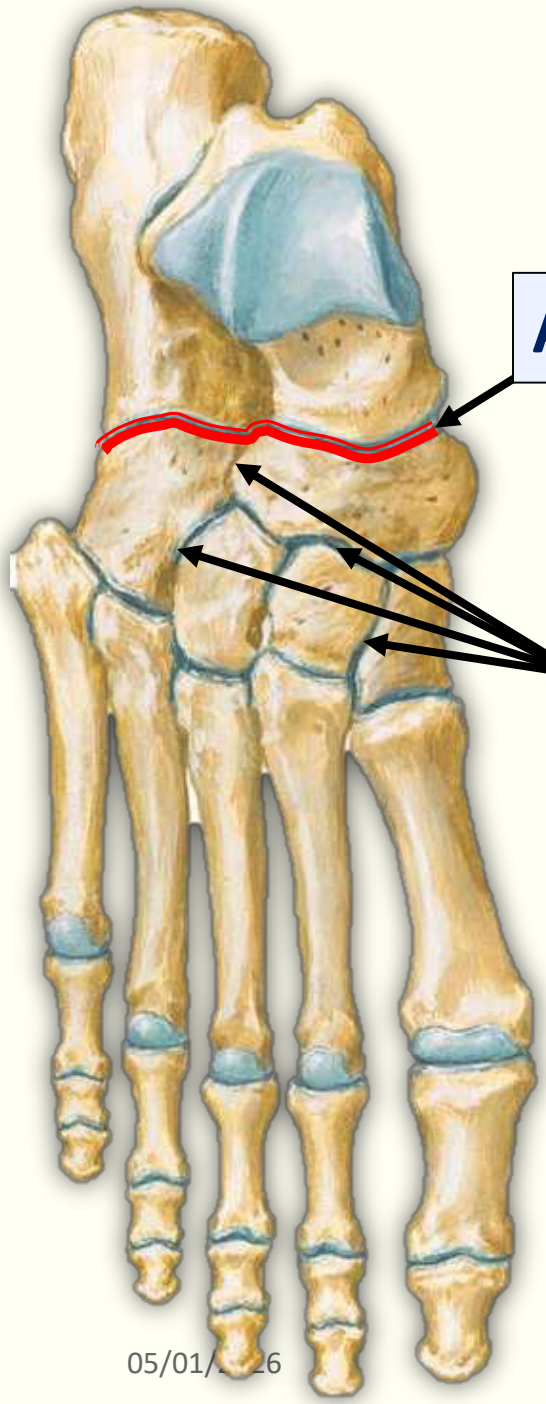
- Ou: **cheville**
- = **diarthrose**, type **ginglyme**
- Relie:
  - **2 os de la jambe**
  - **Talus**

- **Articulation encastrée**
- Assure:
  stabilité du pied
  lors des mouvements
  (**marche** et **course**)





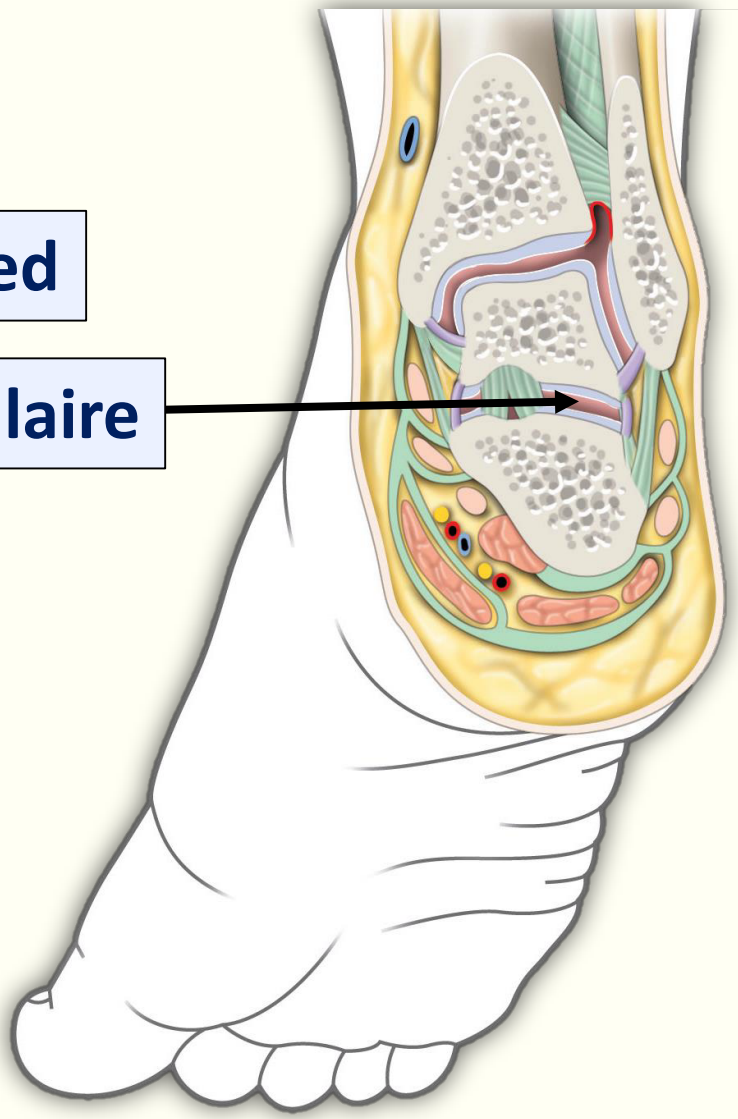
(2)
Articulations du tarse



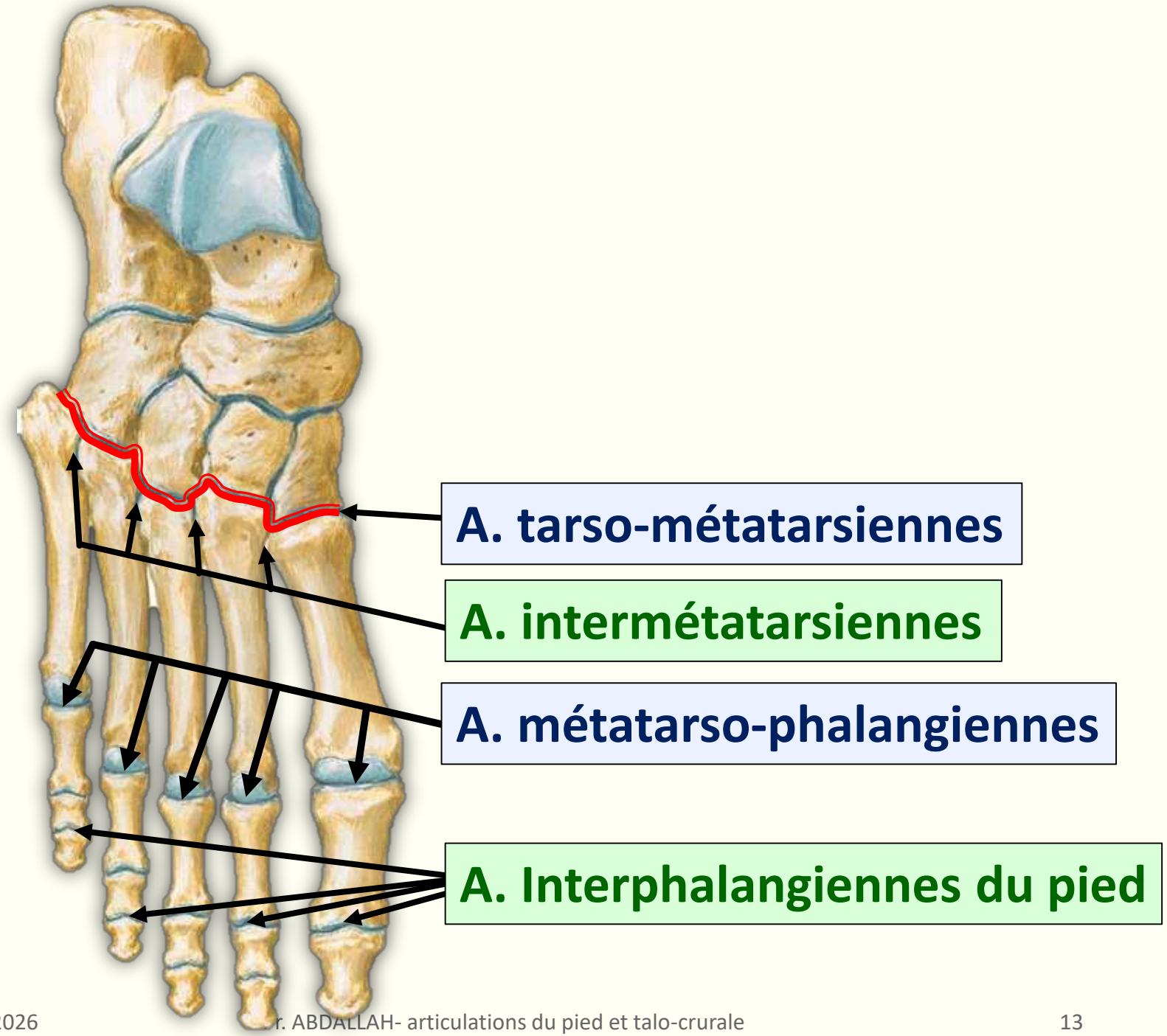
A. transverse du pied

A. subtalaire

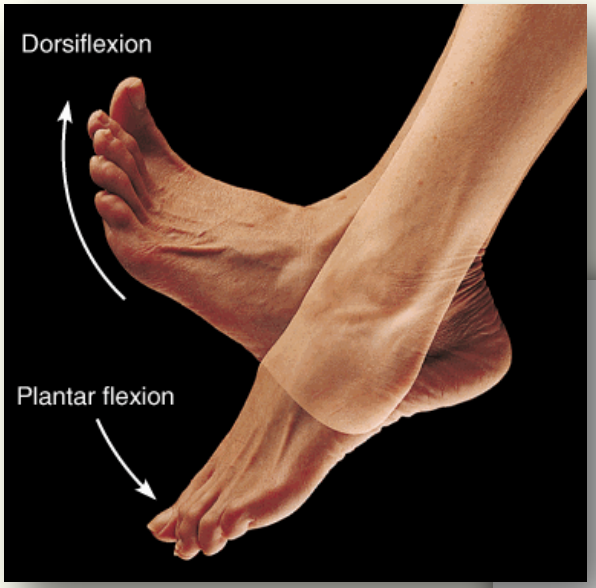
A. des os du tarse distal entre eux

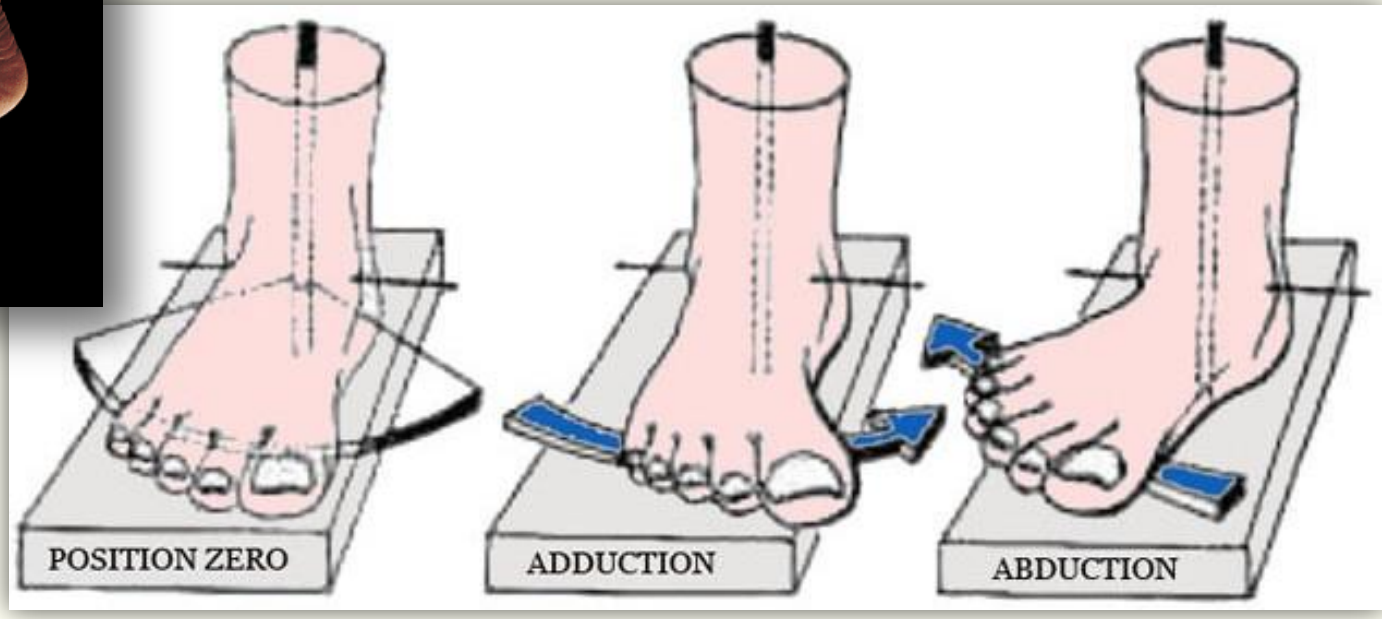


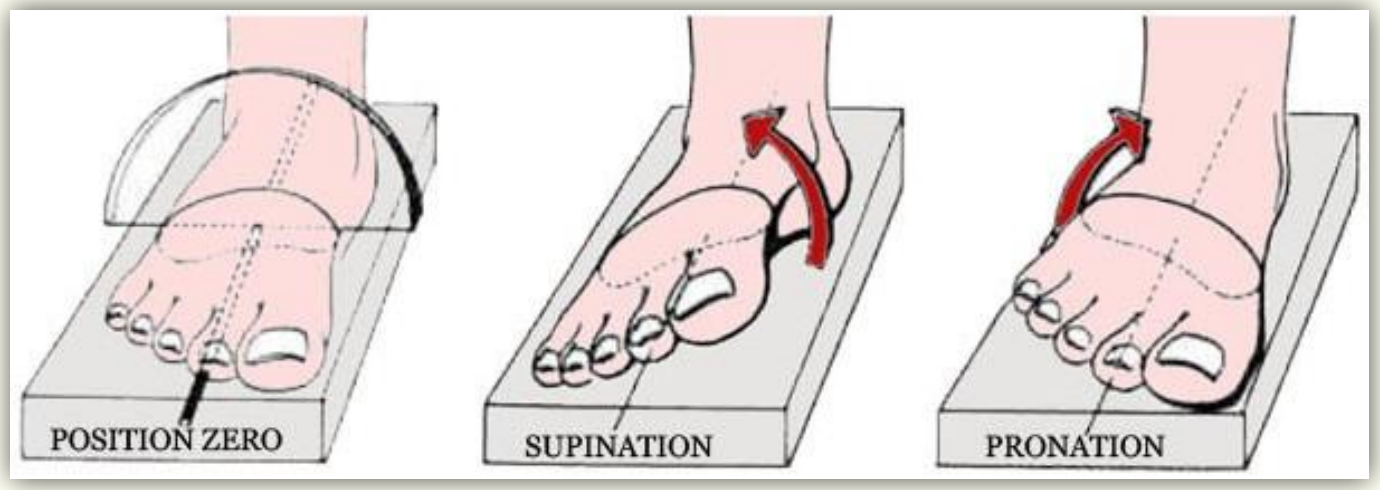
(3)
Articulations de l’avant-pied

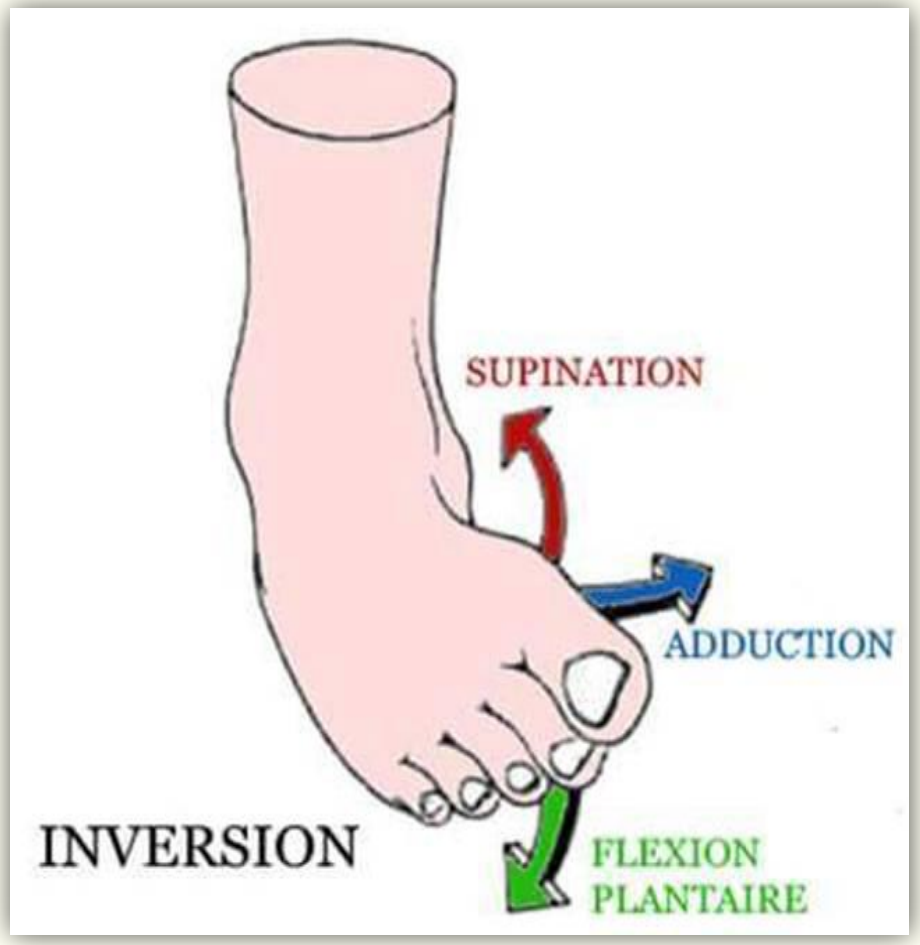

A. tarso-métatarsiennes
A. intermétatarsiennes
A. métatarso-phalangiennes
A. Interphalangiennes du pied

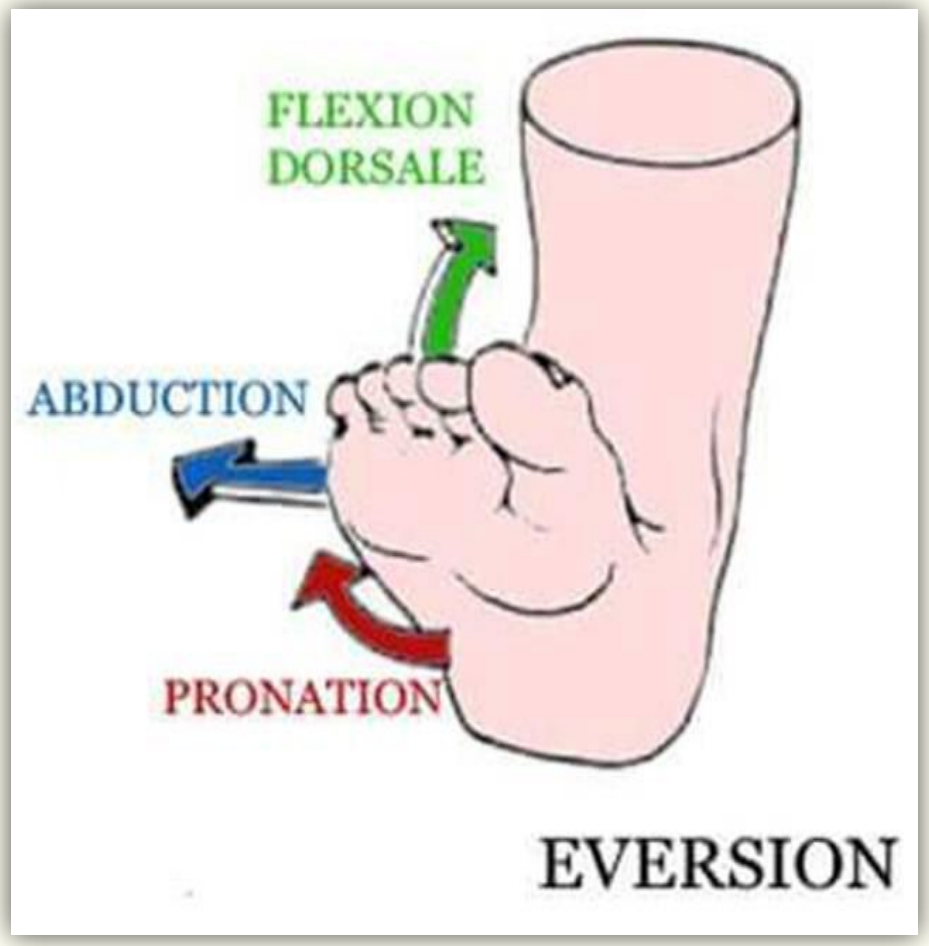
Mouvements du pied



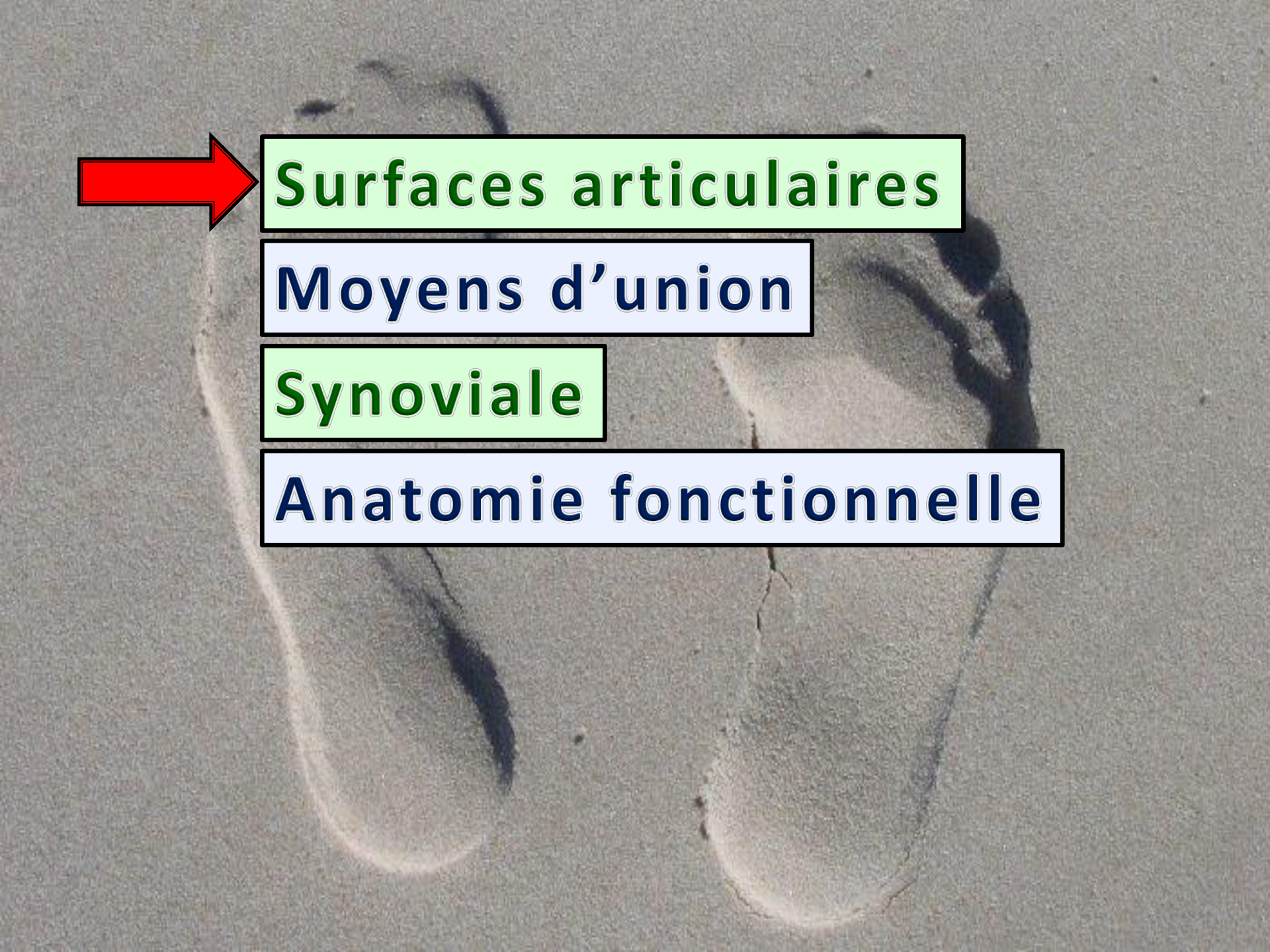





SUPINATION
ADDUCTION
INVERSION
FLEXION
PLANTAIRE

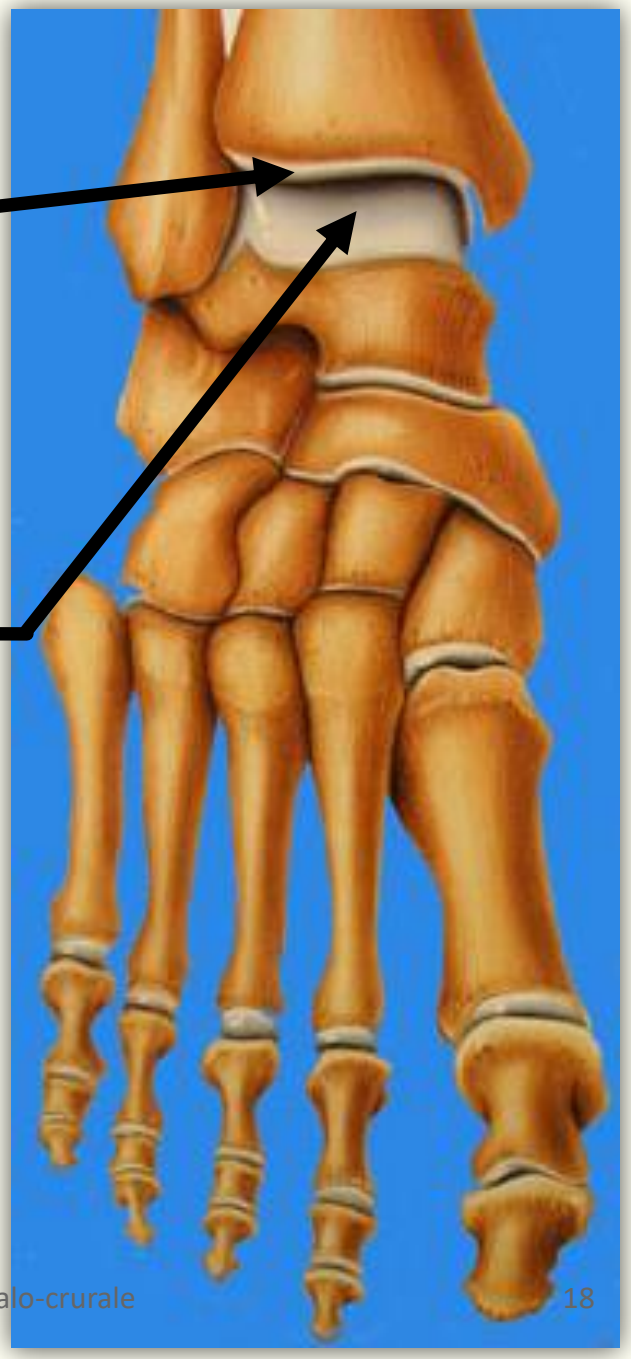

FLEXION
DORSALE
ABDUCTION
PRONATION
EVERSION


Articulation
talo-crurale

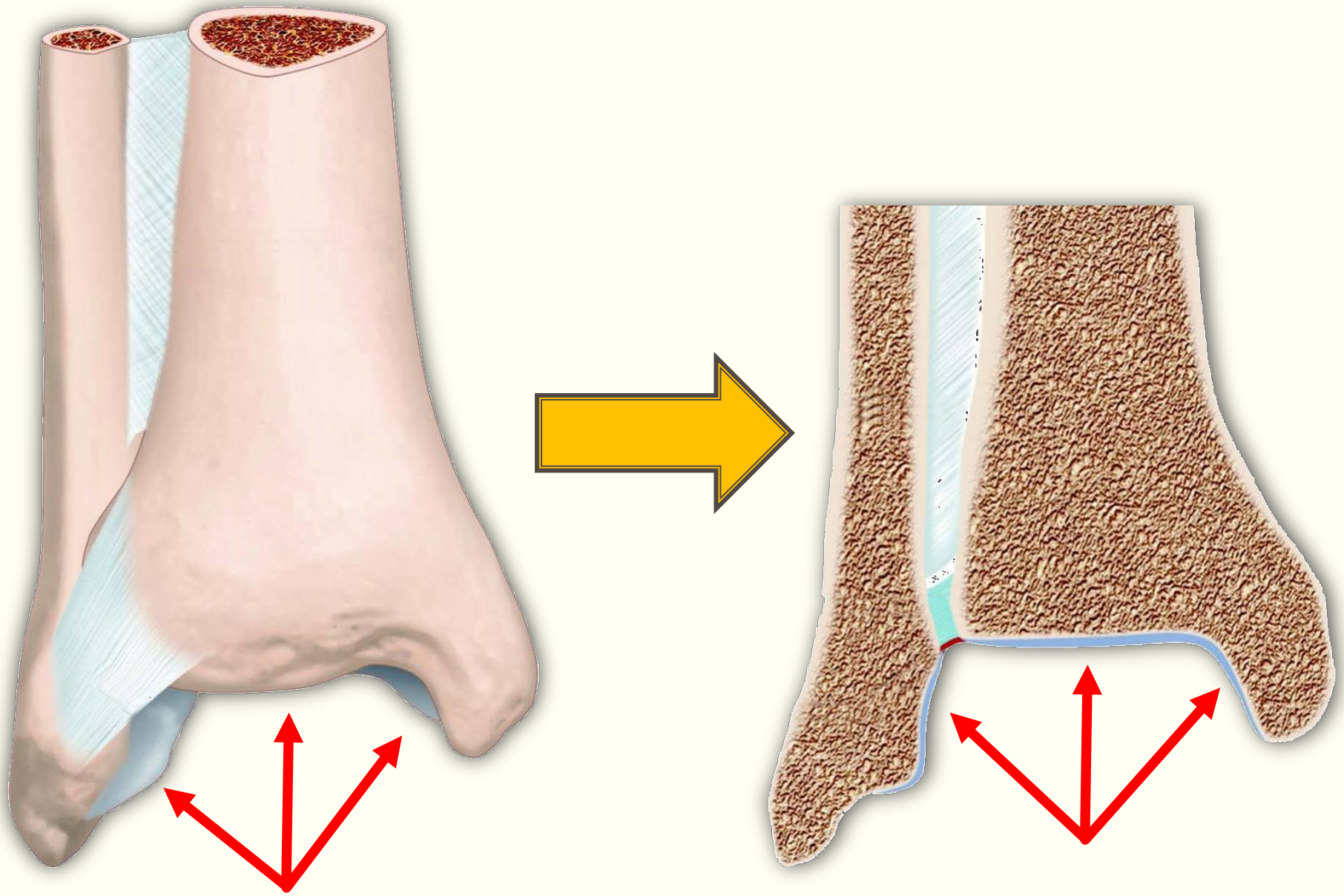

Surfaces articulaires
Moyens d’union
Synoviale
Anatomie fonctionnelle

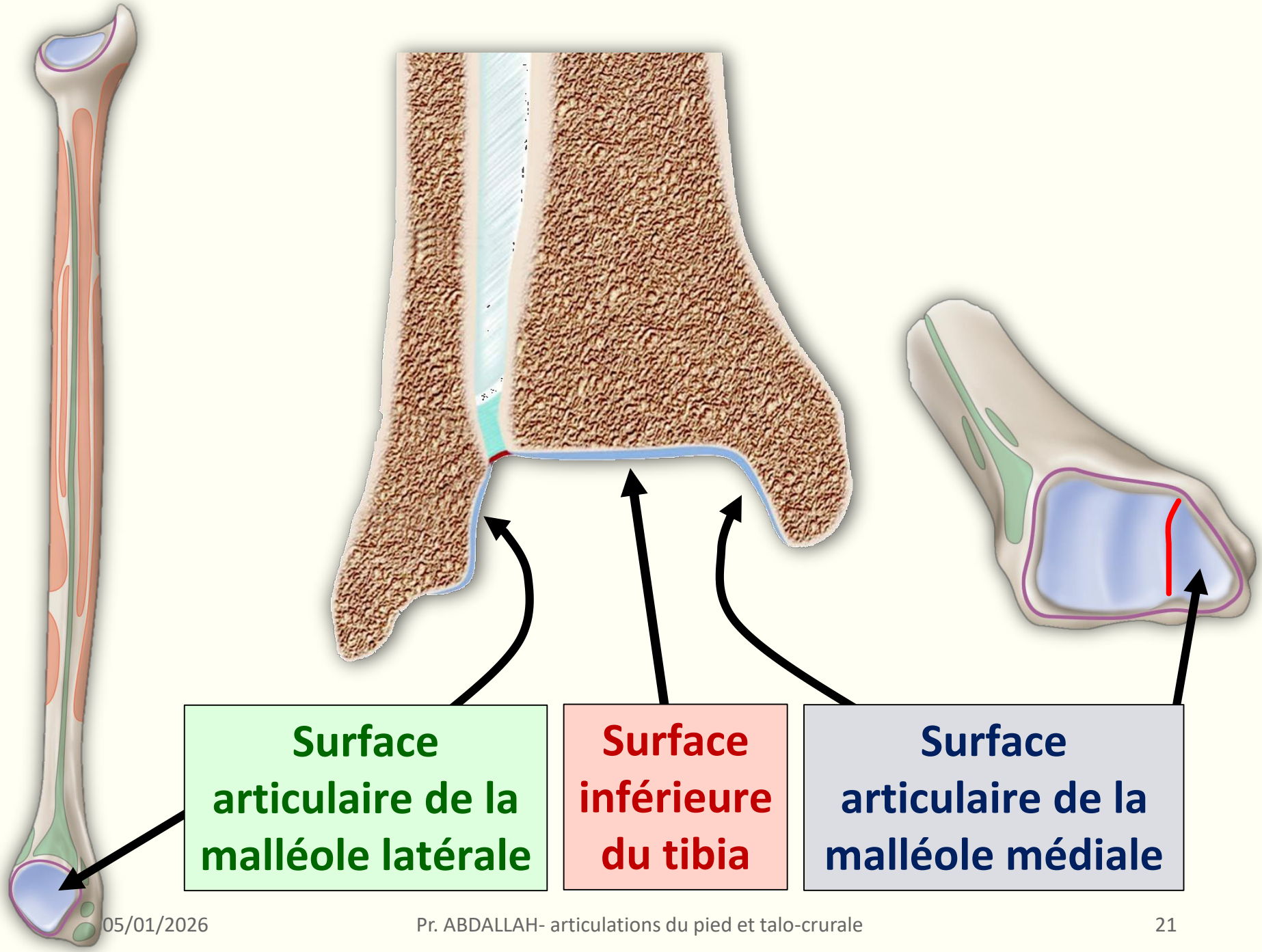
3 surfaces articulaires tibio-fibulaires

3 surfaces articulaires de la trochlée du talus



Surfaces articulaires tibio-fibulaires

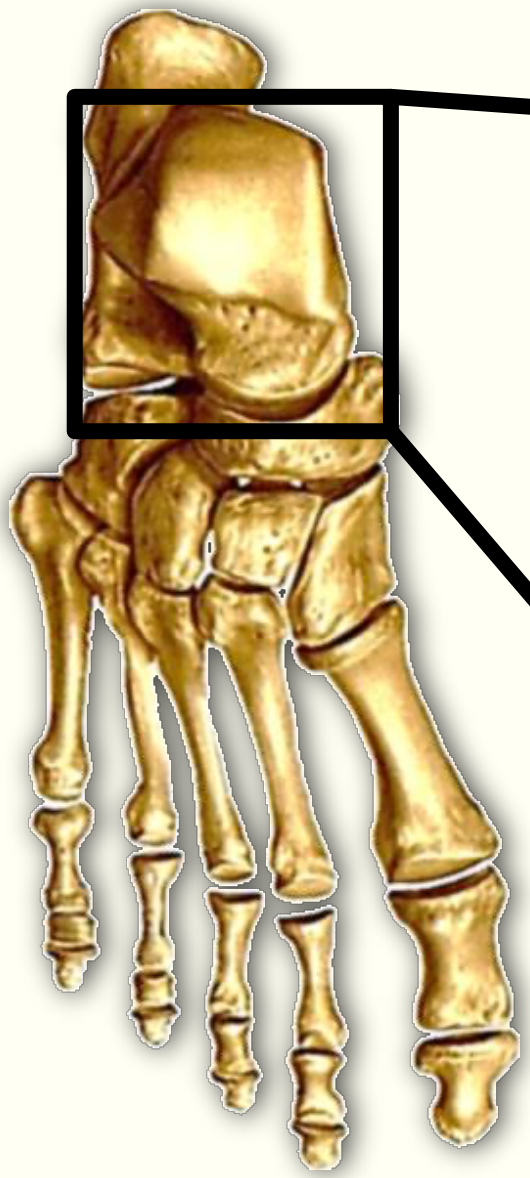


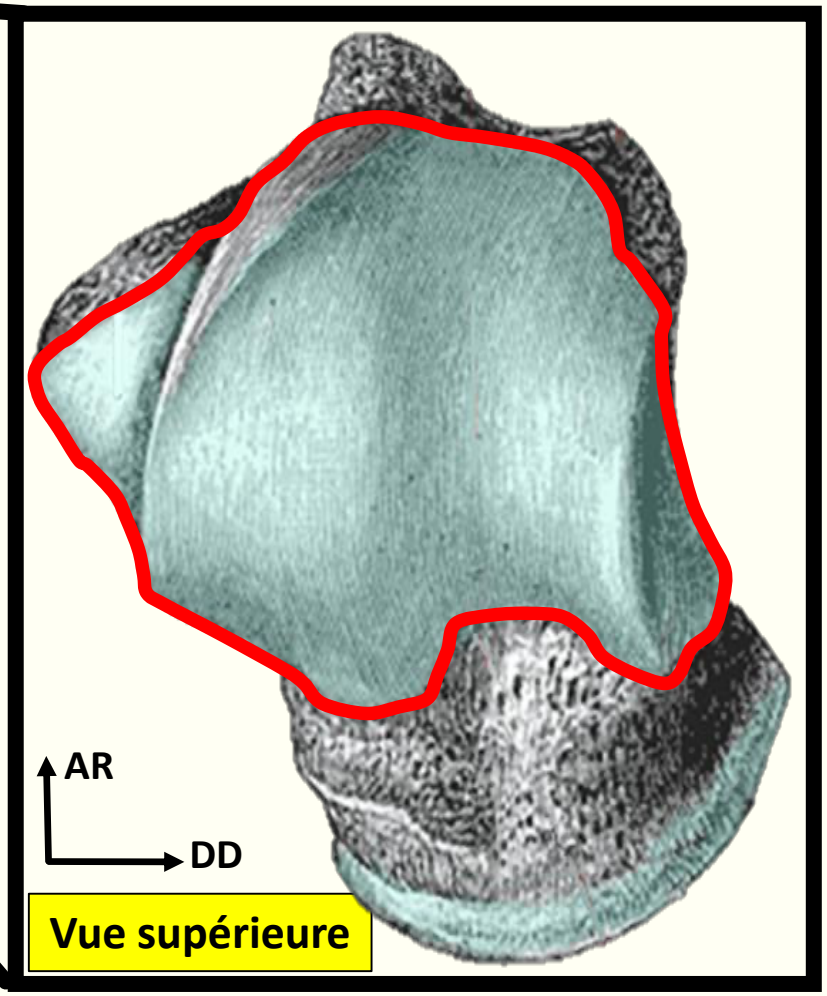

Surface articulaire de la malléole latérale
Surface inférieure du tibia
Surface articulaire de la malléole médiale

Trochlée du talus

= **volumineuse saillie articulaire**

Présente: **3 surfaces…**





articulaire avec:
surface inférieure du tibia

Surface supérieure de la trochlée

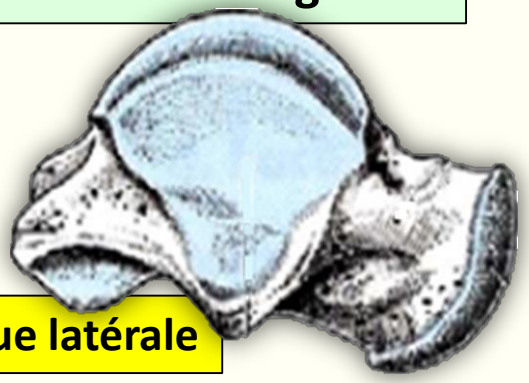
AR

DD

Surface malléolaire latérale

articulaire avec:
malléole latérale

Forme = **triangulaire**



Vue latérale

Surface malléolaire médiale

articulaire avec:
malléole médiale

Forme = **virgule**



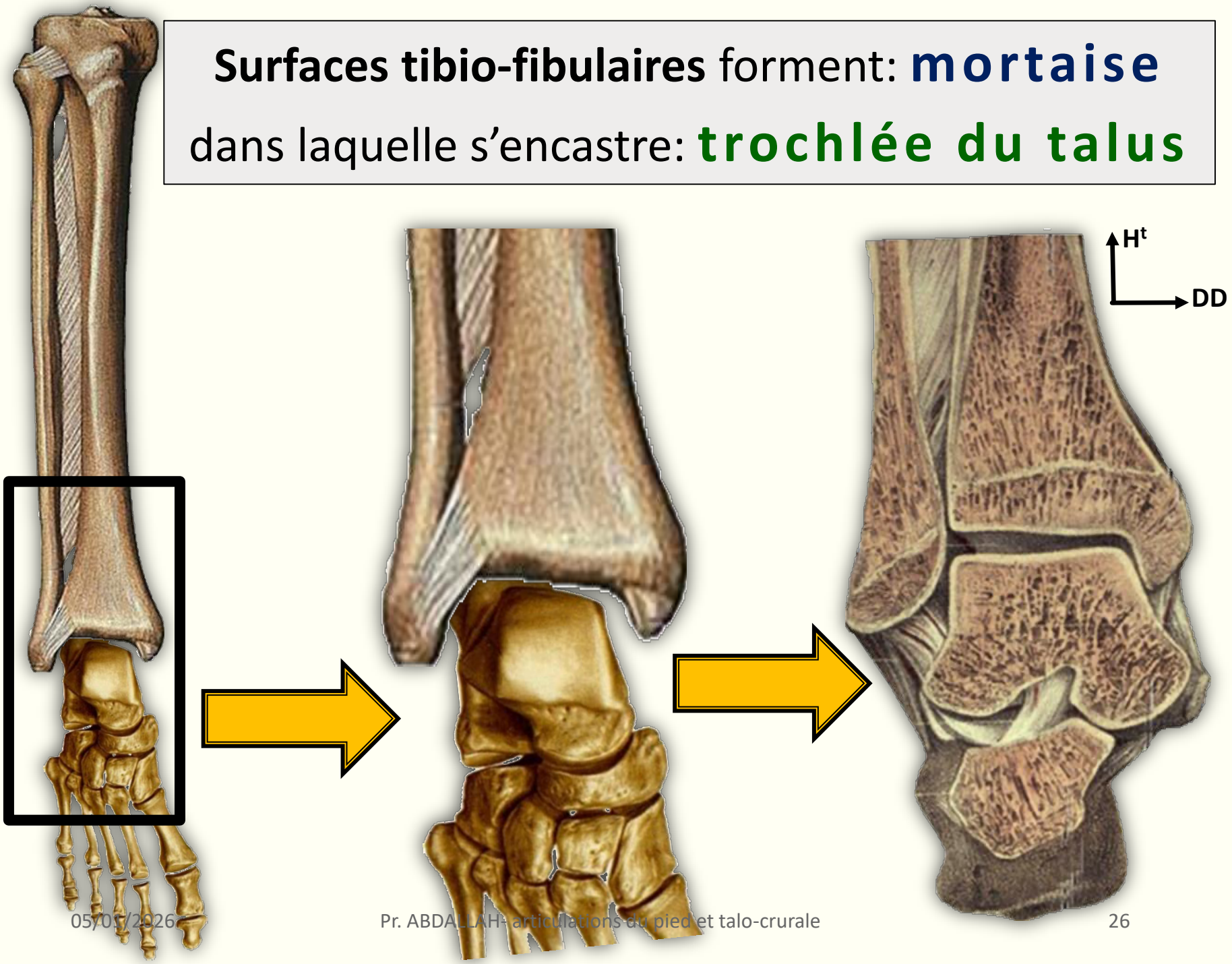
Vue médiale

Comment toutes ces **surfaces articulaires** **s'organisent**



Surfaces tibio-fibulaires forment: **mortaise**

dans laquelle s'encastre: **trochlée du talus**



Surfaces articulaires

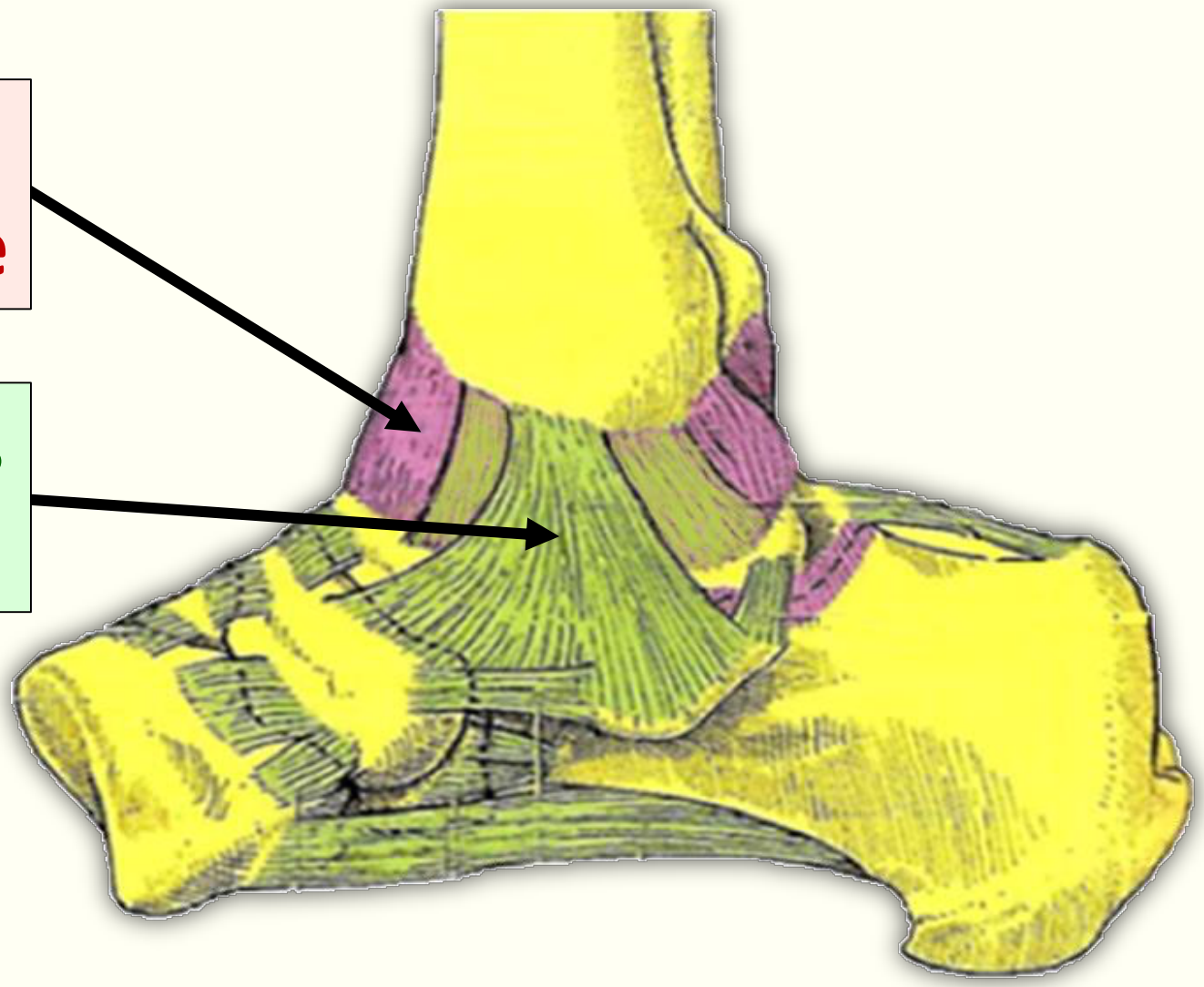
Moyens d’union

Synoviale

Anatomie fonctionnelle

Capsule articulaire

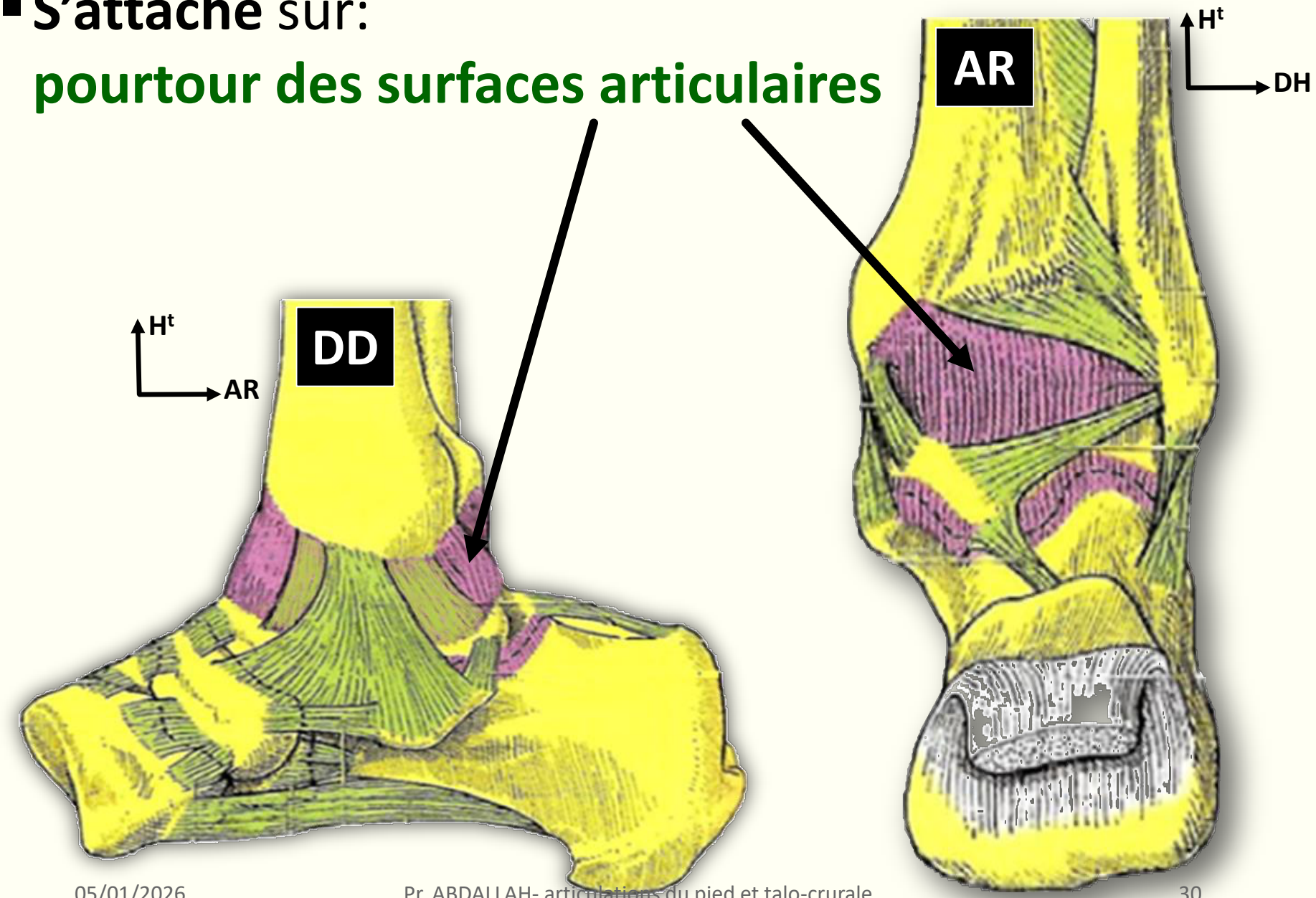
Ligaments passifs



Capsule articulaire

- **S'attache** sur:

  pourtour des surfaces articulaires

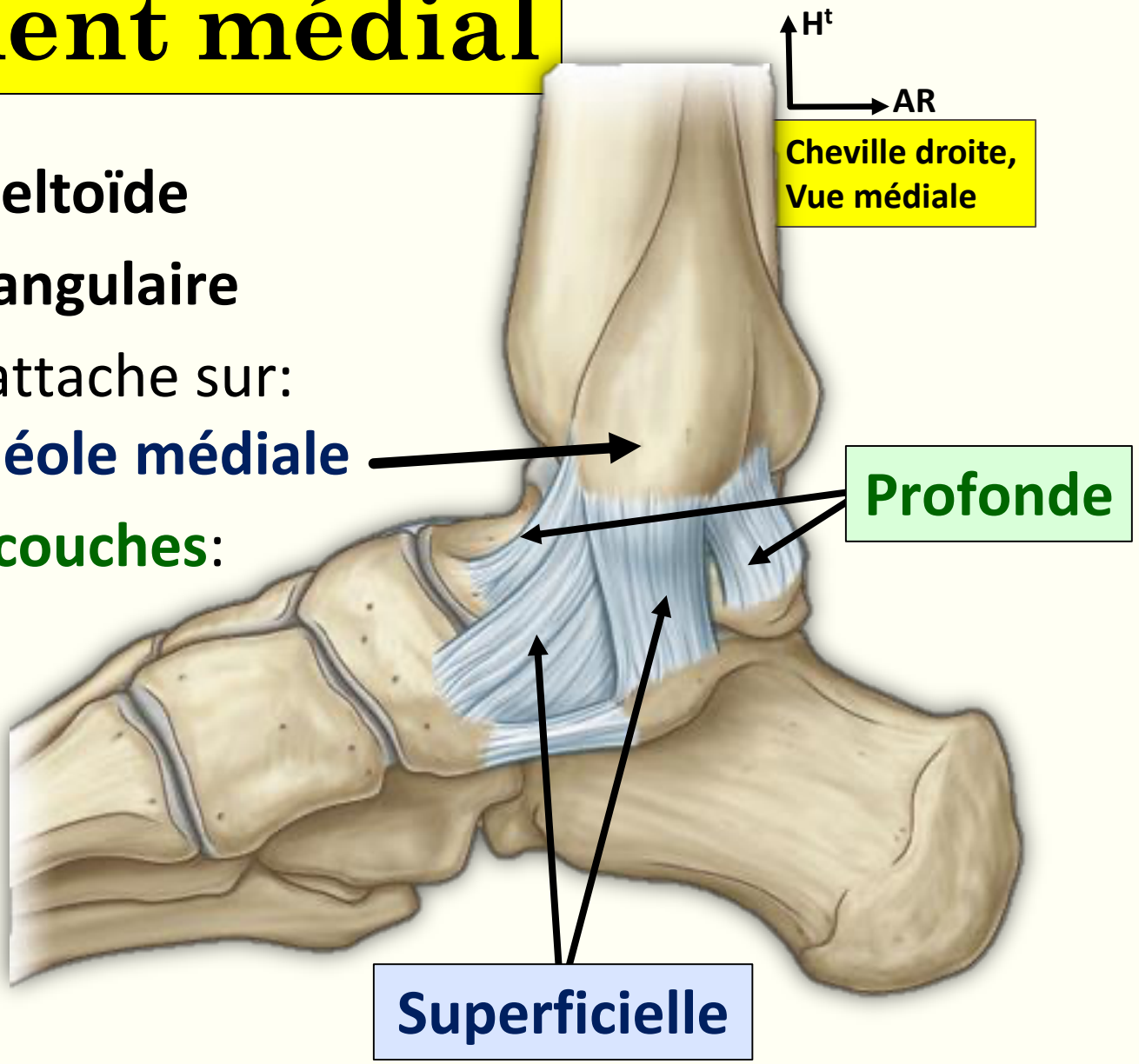


Ligaments passifs

Au nombre de 2

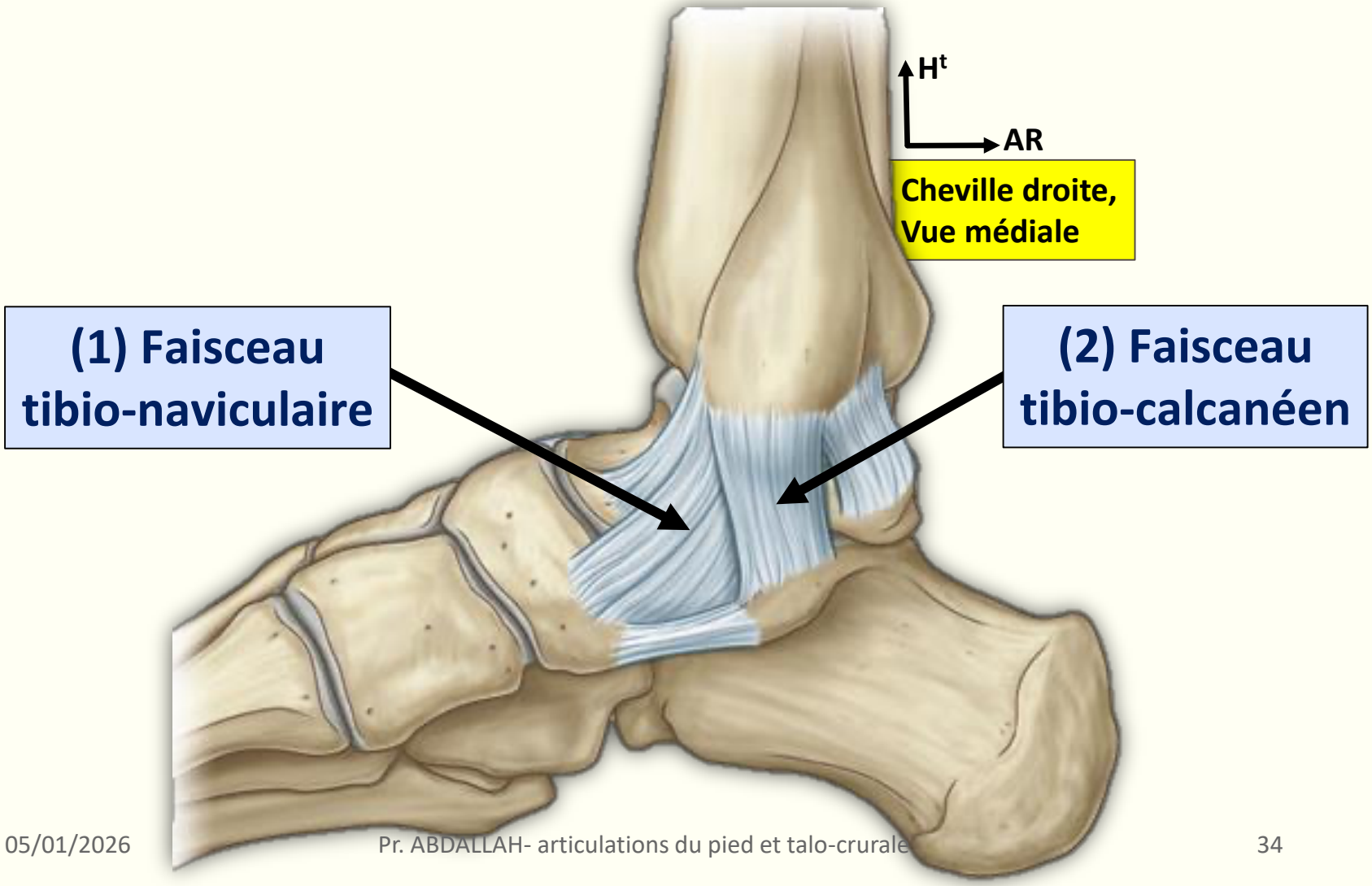
1 Ligament médial

- Ou: **Ligament deltoïde**
- **Résistant** et **triangulaire**
- Son sommet s'attache sur:
  malléole médiale
- Constitué de **2 couches**:



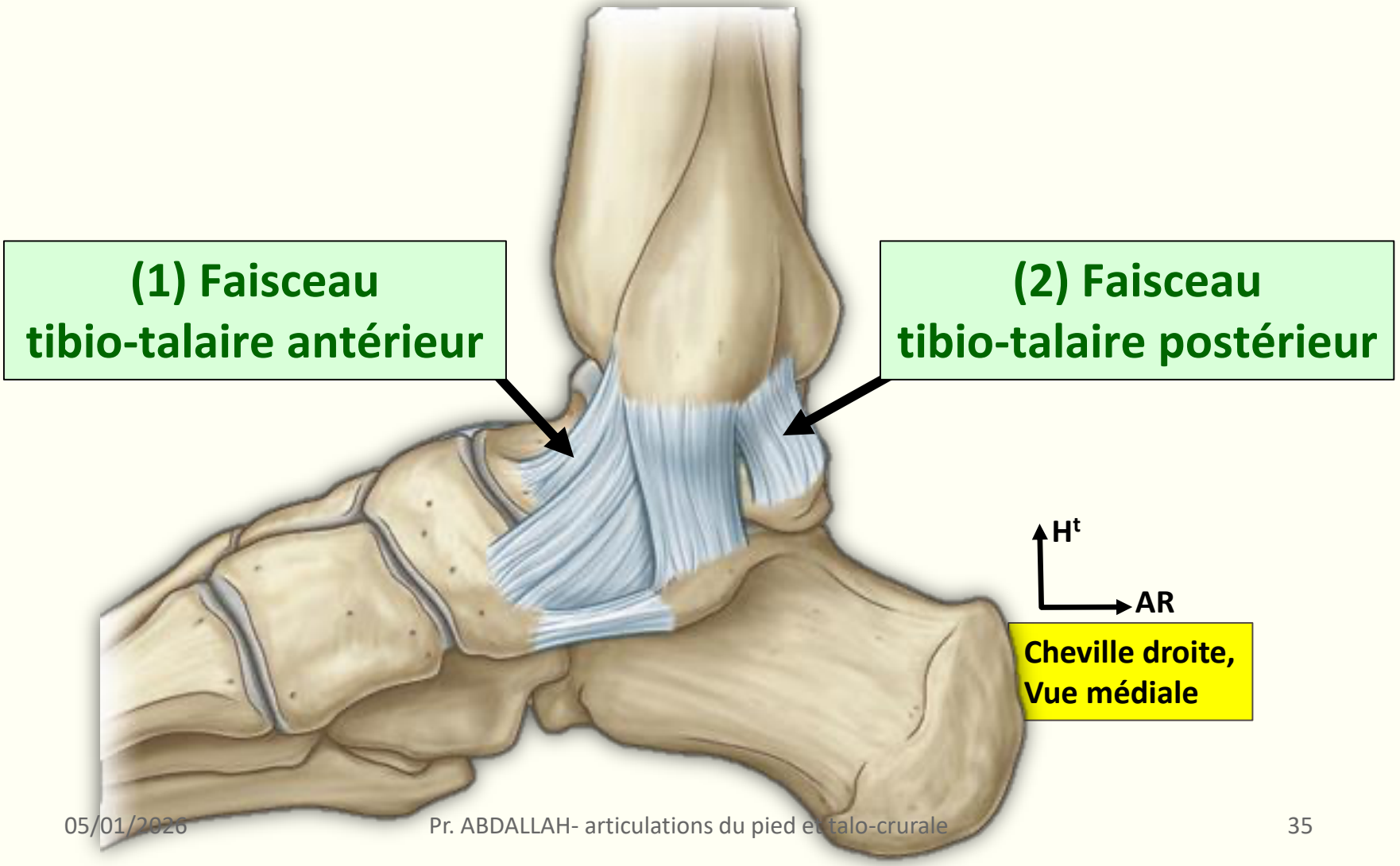
Couche superficielle

Comprend d'**avant** en **arrière**:



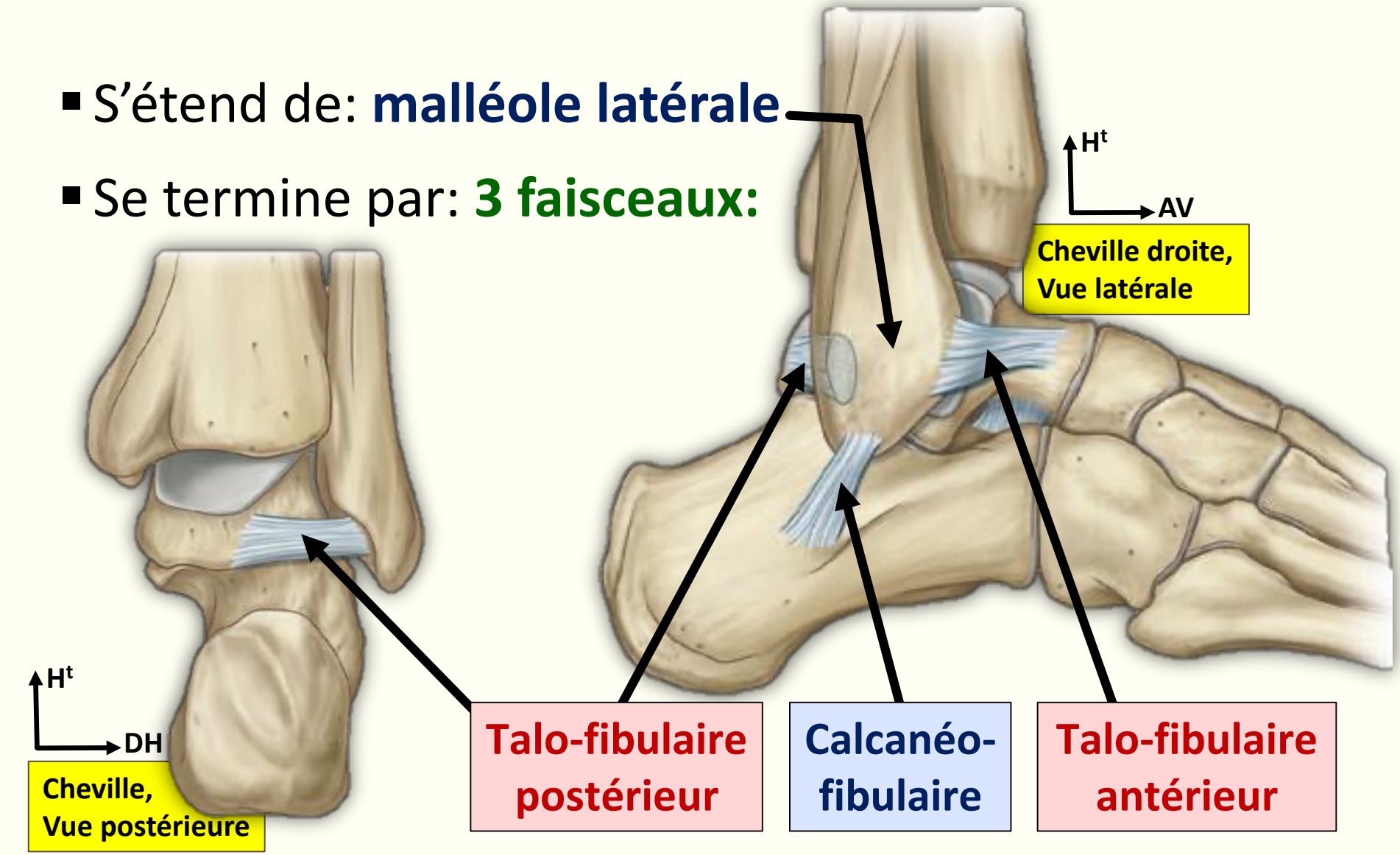
Couche profonde

Comprend d'**avant** en **arrière**:



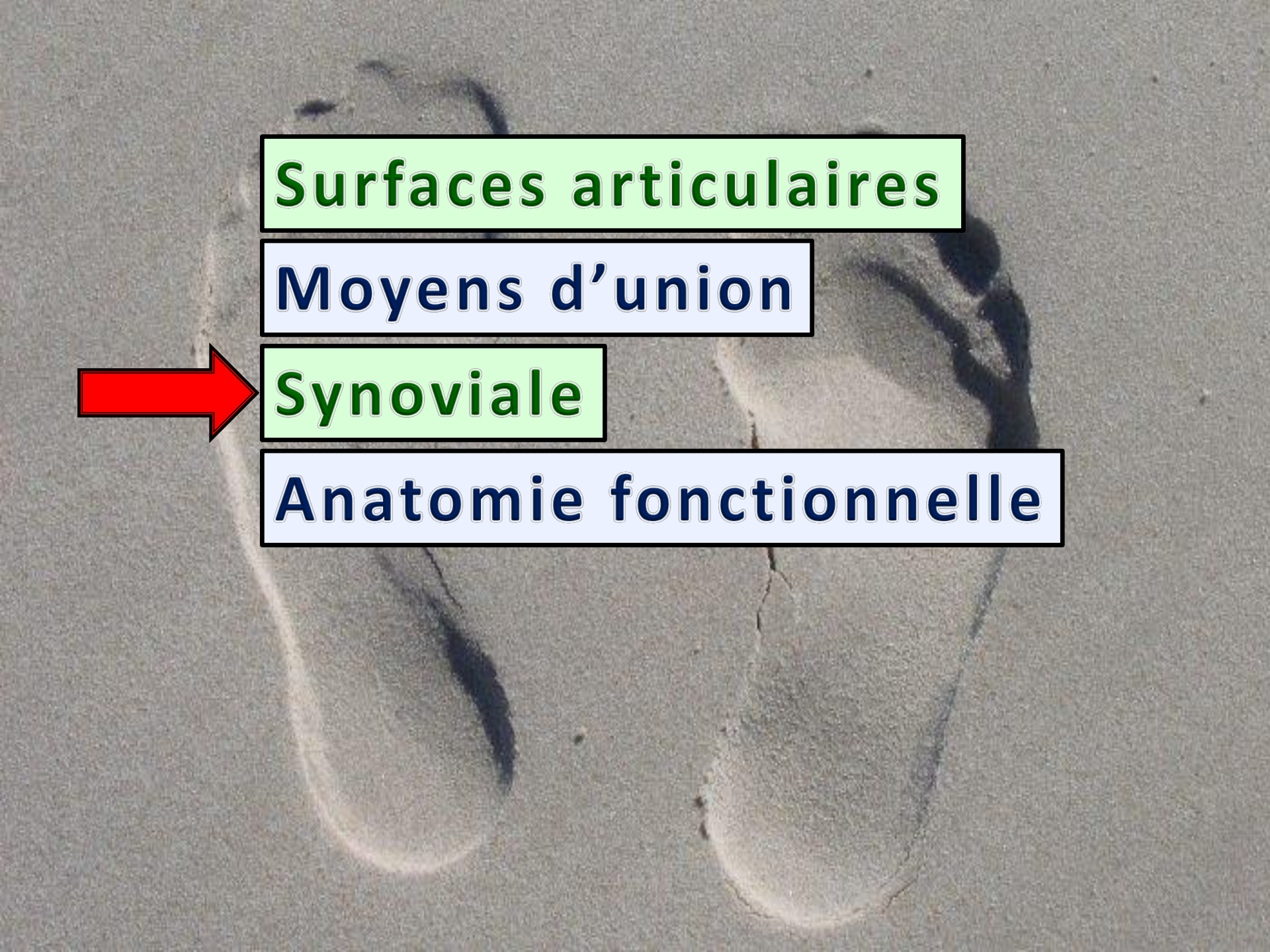
2 Ligament latéral

- S'étend de: **malléole latérale**
- Se termine par: **3 faisceaux:**



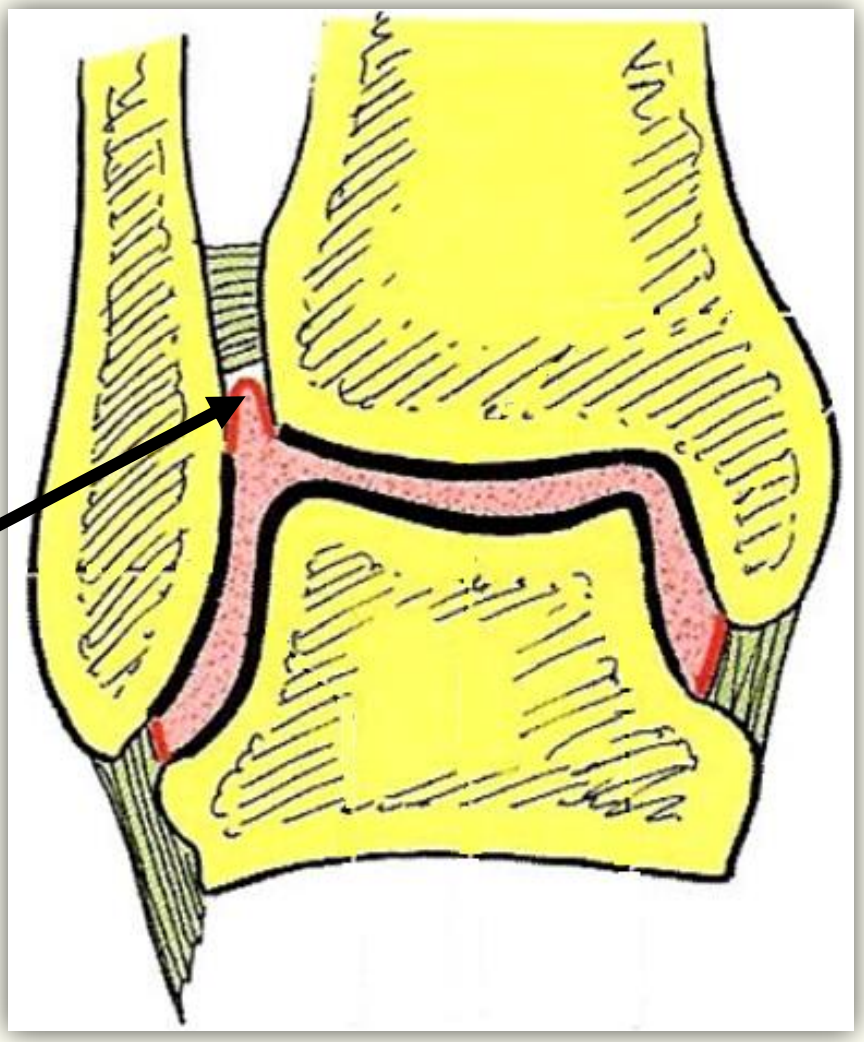
Ligament latéral est
« le ligament de l'entorse de la cheville »

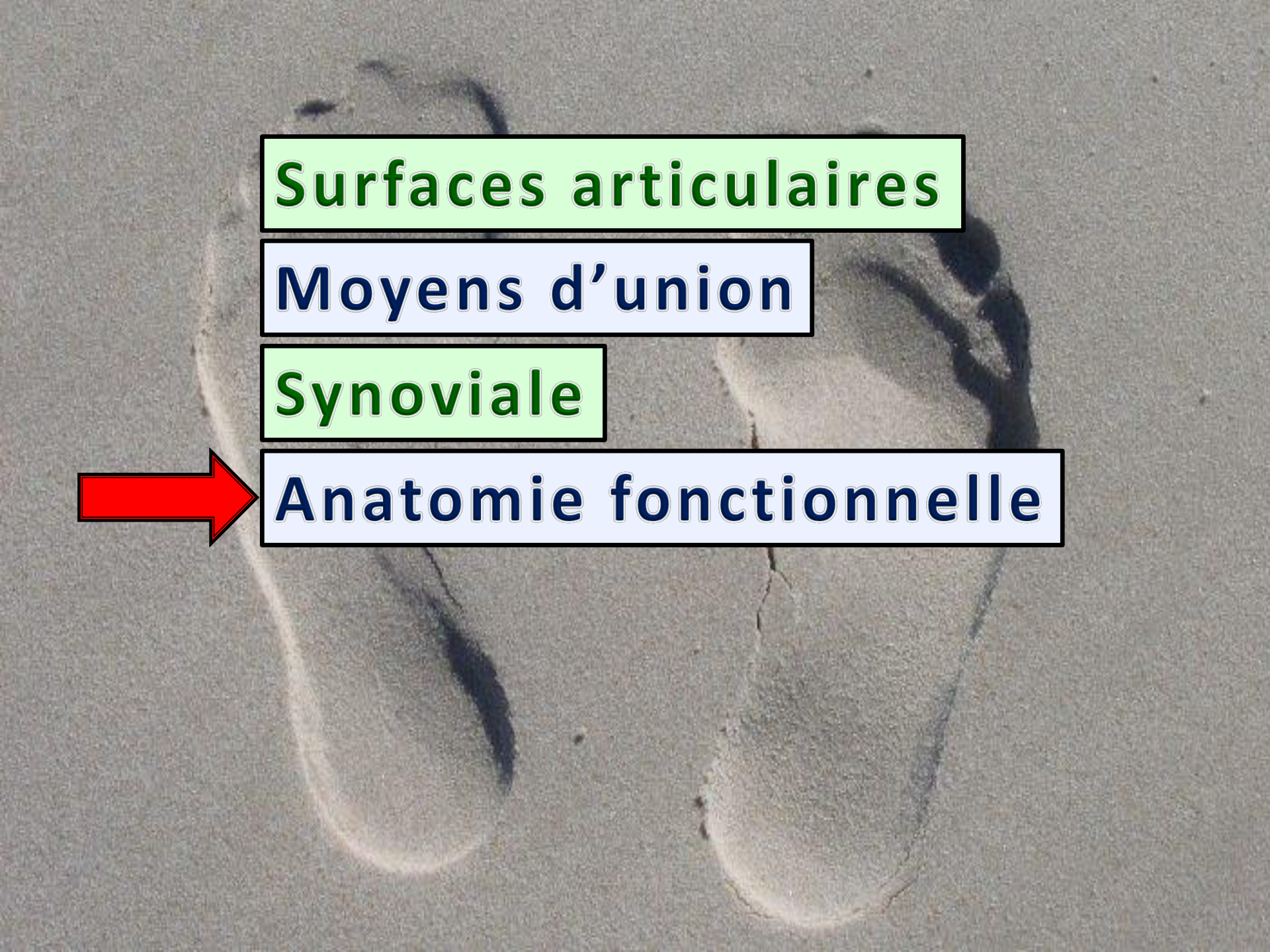



Surfaces articulaires
Moyens d’union
Synoviale
Anatomie fonctionnelle

- **Tapisse:**
  face profonde
  de la capsule

- **Donne:**
  prolongement
  en **cul-de-sac**
  entre:
  tibia et **fibula**



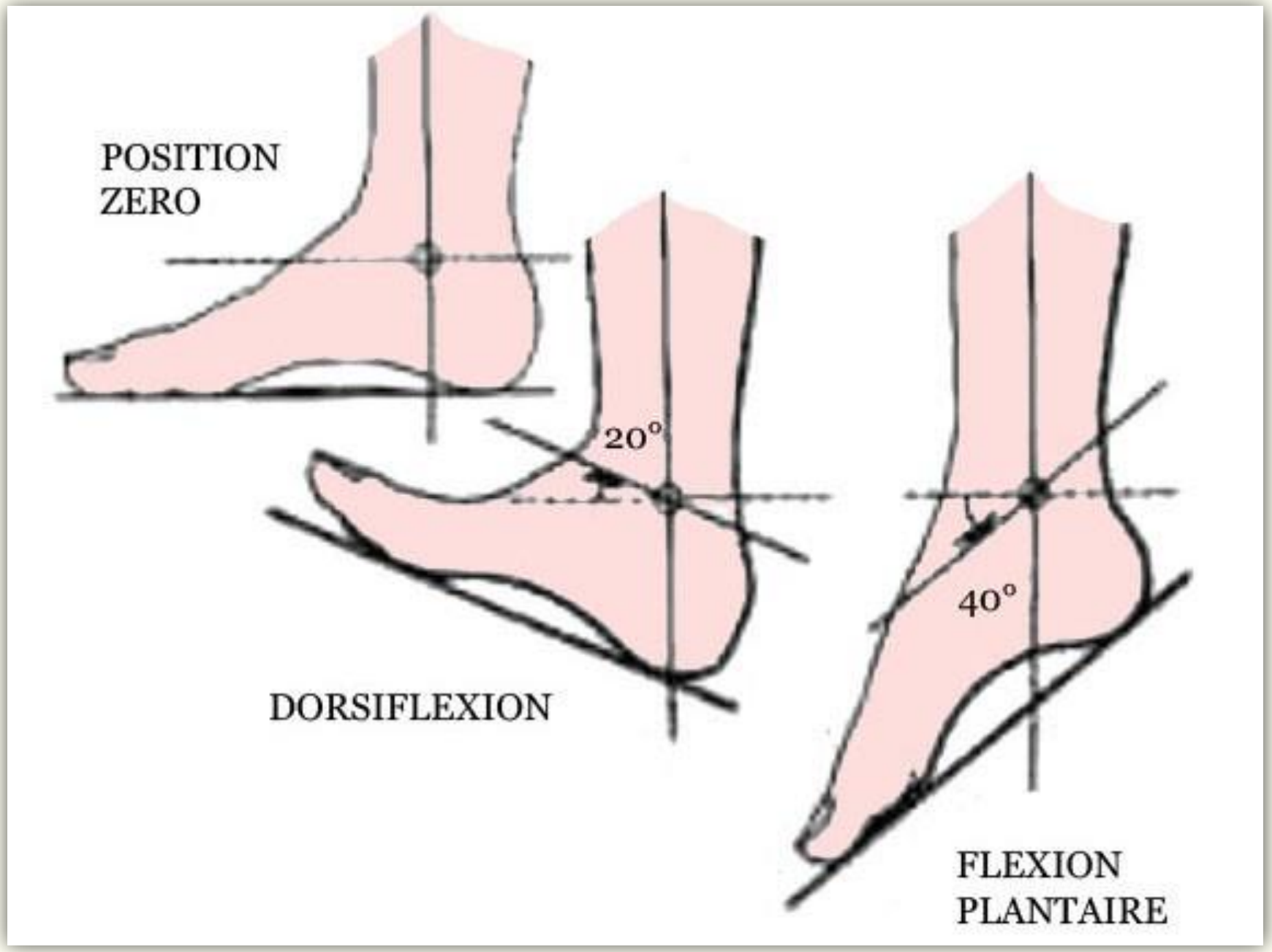

Surfaces articulaires
Moyens d’union
Synoviale
Anatomie fonctionnelle

- Présente:
  un seul degré de liberté

- Permet:
  flexion-extension du pied

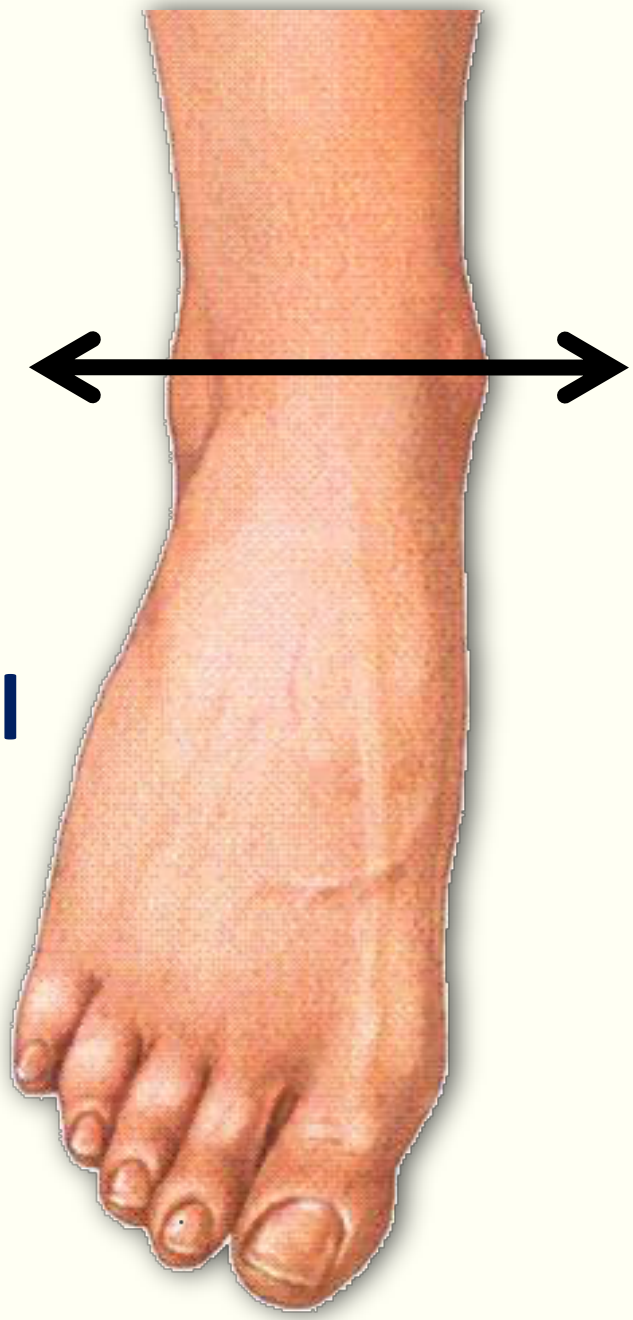
  (= **élévation-abaissement** de l'avant pied)




POSITION
ZERO
20°
DORSIFLEXION
40°
FLEXION
PLANTAIRE

- Se font autour

  d’un **axe transversal**



Flexion

- **Rapproche:**
  face dorsale du pied
  à **la face antérieure**
  de la jambe
- **Amplitude = 20° à 30°**





Extension

- **Éloigne:**
  face dorsale du pied
  de **la face antérieure**
  de la jambe

- **Amplitude = 30° à 60°**





Bonne révision

